

Baker's Legacy

1. Project Overview

It is a simple 3D OpenGL game where the player, after going hilariously bankrupt, is forced to work at a cake shop to earn enough money to recover financially. The player receives customer orders and must prepare cakes by combining simple geometric shapes.

The cakes can be **square, rectangular, or round of different colours**, with either one or two tiers and an optional topping. While serving customers, random robbers may enter the shop and demand money. The player can deal with them by calling the police, playing Rock-Paper-Scissors, or taking a chance with a cake-eating competition.

The ultimate objective is to reach a randomly generated **target amount of money** while managing orders, robber encounters, and limited police calls. A high-score system adds replayability.

2. Core Features

2.1 The Cake Shop

The game takes place inside a simple 3D cake shop containing the cake-making counter, customer area, cash counter, and other basic shop objects. The environment is built using simple OpenGL geometric primitives such as cubes, cuboids, and cylinders.

2.2 Customer Orders

Customers arrive at the shop and provide a specific cake order. Each order contains requirements such as:

- Cake shape
- Cake flavour (Vanilla, Chocolate, Strawberry)
- Number of tiers
- Topping preference

The player must carefully follow the order and submit the completed cake to receive payment.

2.3 Three Cake Shapes & Flavour

The player can make three different shapes and 3 different flavours of cakes:

Shapes:

- **Square** — represented by a cube.
- **Rectangle** — represented by a rectangular cuboid.
- **Round** — represented by a cylinder.

Colour:

- **Chocolate** — Brown coloured shape.
- **Vanilla** — Ivory coloured shape.
- **Strawberry** — Pink coloured shape.

The required shape and flavour are determined by the customer's order.

2.4 Cake Tier System

Each cake can contain either **one tier or two tiers**.

A two-tier cake is created by placing a smaller cake shape on top of a larger cake shape. The upper layer must be smaller than the bottom layer to create the appearance of a tiered cake.

2.5 Optional Toppings

There are two available toppings:

- **Cherry** — represented by a small sphere.
- **Chocolate Bar** — represented by a small cuboid.

Customers may have a topping preference, but toppings are optional. Only **one topping can be used on a cake at a time**.

2.6 Cake Assembly

The player must assemble the cake according to the current order. The basic process is:

1. Select the required cake shape.
2. Select the number of tiers.
3. Add the requested topping if required.
4. Submit the completed cake.
5. Receive money if the order is correct.

Incorrect cakes can result in no payment or a penalty.

2.7 Money and Target System

Every successfully completed order increases the player's money. At the beginning of the game, a **random target amount** is generated.

For example:

Money: \$320

Target: \$500

The player must continue completing orders and managing robber attacks until the current money reaches or exceeds the target.

2.8 Robber Events

Randomly during gameplay, a robber can enter the cake shop and demand money from the player.

The player must immediately decide how to handle the situation (a timeline will be given. The player has to make a decision within the limited time. Otherwise, the robber takes a portion of money and leaves.) . The robber encounter temporarily interrupts the normal cake-making gameplay and introduces an element of risk and randomness.

2.9 Police Calls

The player starts with a limited number of **3 police calls**.

Calling the police causes the robber to leave, but one police call is consumed.

Police Calls: 2 / 3

The player can regain police calls by achieving certain financial milestones or bonus targets, encouraging them to keep earning money.

2.10 Rock-Paper-Scissors

Instead of using a police call, the player can challenge the robber to **Rock-Paper-Scissors**.

The player chooses:

- Rock
- Paper
- Scissors

The robber makes a random choice. If the player wins, the robber leaves. If the player loses, the robber can steal money or apply another penalty.

A draw can result in another round.

2.11 Cake eating game

The player can alternatively challenge the robber to a **cake-eating game**.

The game angle would change. The player and robber will both have cakes in front of them, and whoever finishes the cake first will win. During the cake game, both the player and robber can shoot whipped cream on each other it will increase the size of the cake, and they would need to eat more.

- **Player Wins** — Robber has to leave with nothing, and the game continues.
- **Robber Wins** — The robber will take the player's Entire income and leave. The game continues.

But either way, a certain amount of coins will be taken away from the player as the price of the cakes.

2.12 Robber Money Loss

If the player fails to stop a robber, the robber can steal a portion of the player's money.

For example:

Robber stole \$50!

Money: \$270

Target: \$500

This makes robber encounters important because losing money increases the amount of work required to reach the target.

2.13 Random Coin Collection

Sometimes the player may find random coins lying around, left by the customer. The player can click on them to collect them. Those coins will be added to the player's income.

2.14 Financial Milestones

The player can receive rewards after reaching certain money milestones.

Possible rewards include:

- Additional police calls
- Bonus money
- Other shop rewards

These milestones give the player additional goals while working toward the final target.

2.14 Simple 3D OpenGL Objects

The game uses basic 3D primitives to construct the gameplay objects.

Object	Representation
--------	----------------

Square Cake	Cube
Rectangular Cake	Cuboid
Round Cake	Cylinder
Cherry	Sphere
Chocolate Bar	Cuboid
Shop Objects	Cubes / Cuboids
Customer	Simple 3D model
Robber	Simple 3D model

This keeps the game visually simple while allowing the project to demonstrate 3D transformations, object placement, animation, collision detection, and user interaction.

3. Controls

Key	Used for	Action
1/2/3	Decision making	Select among 3 tiles for the desired outcome.
P	Pausing/playing the game	Click on “P” on the keyboard.
→	Skipping conversation	Click on “→” on the keyboard.
E	Exit the game	Click on “E” on the keyboard.
Mouse	Coin collection	Navigate/collect random coins lying around.
↑	Throwing whipped cream	During the cake-eating game, the “↑” key is used to throw whipped cream.
↓	Eating the cake	During the cake-eating game, the “↓” key is used to eat the cake.
X	Refusing to serve a Customer	Click on “X” on the keyboard.
A	Changing the cake shop angle	Click on “A” on the keyboard.

3. Overall Objective

The main objective of Cake Shop is to turn a simple cake-making mechanic into a humorous survival-and-management game.

The player must balance three things:

1. Making the correct cakes to earn money.
2. Managing police calls when robbers appear.
3. Taking risks with Rock-Paper-Scissors and the cake-eating game.

The combination of simple OpenGL geometry, customer orders, random targets, robber encounters, and mini-games creates a lightweight but replayable game suitable for an OpenGL project.